

Conditional formatting is a powerful visual tool in Excel that automatically changes the appearance of cells based on specific rules or conditions you set. It allows you to highlight patterns, identify errors, and emphasize important data points without changing the actual data.

## 1. Core Functions and Visual Cues

Conditional formatting provides several ways to visualize data directly within your spreadsheet grid:

- **Highlighting Duplicates:** One of the most common uses is to identify and remove duplicate entries in a list to ensure data integrity.
- **Color Scales:** Applies a gradient of colors across a range of cells to represent the magnitude of values (e.g., green for high values and red for low values).
- **Data Bars:** Inserts a horizontal bar inside the cell, where the length of the bar represents the value relative to other cells in the range.
- **Icon Sets:** Places small icons (like arrows, traffic lights, or checkmarks) next to values to indicate their status or trend.

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## 2. Logical Rules for Formatting

You can create custom rules using the same logical operators found in the **IF function**:

- **Greater than / Less than:** Highlight salaries above a certain threshold or scores below a passing mark.
- **Text that Contains:** Format cells that contain specific keywords, such as "Pass," "Fail," or a specific city like "Hubli".
- **Dates Occurring:** Highlight deadlines that are approaching within the next week or month.

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## 3. Advanced Applications

Beyond simple highlights, conditional formatting can be used for sophisticated data management:

- **Error Detection:** Use rules to highlight cells containing errors (like #N/A or #VALUE!) or values that violate **Data Validation** rules.
- **Dynamic Highlighting:** You can write a formula as a rule so that formatting updates automatically whenever you change the data.
- **Search Results:** Highlight a whole row based on a search term entered in a specific cell.

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## 4. How to Apply Conditional Formatting

1. **Select the Cells:** Highlight the range you want to format.
2. **Access the Tool:** Go to the **Home tab** and click on **Conditional Formatting**.
3. **Choose a Rule Type:** Select from "Highlight Cells Rules," "Top/Bottom Rules," or create a "New Rule" for custom formulas.
4. **Set the Format:** Choose the fill color, font style, or border that should appear when the condition is met.